

Where irrigation exists is globally contested

3. Crop regridding

Arnald Puy

Contents

1	Session information	5
---	---------------------	---

```

# ARRANGE CROP DATASETS #####
#####

sensobol::load_packages(c("data.table", "terra", "magrittr", "here", "rnatrualearth",
                          "sf", "countrycode"))

# Source all helper functions -----

r_functions <- list.files(path = here("functions"), pattern = "\\R$",
                          full.names = TRUE)

invisible(lapply(r_functions, source))

# Countries layer (WGS84, only name + geometry) -----

countries_sf <- rnaturalearth::ne_countries(scale = "medium",
                                             returnclass = "sf") %>%
  .[, c("name", "geometry")]

# Columns to keep for gridded products -----

cols_to_retrieve <- c("lon", "lat", "country", "continent", "mha", "dataset")

# GAEZ 2015 #####
#####

# LIST ALL IRRIGATED CROP PRODUCTION FILES -----

# Base directory -----
gaez_dir <- "./crop_datasets/GAEZ_2015/GAEZ._2015_crop_production"

# List the irrigated rasters -----

crop_files <- list.files(gaez_dir,
                        pattern = "^GAEZAct2015_Production._*_Irrigated\\.tif$",
                        full.names = TRUE)

# CREATE CLEAN CROP NAMES -----

crop_names_raw <- basename(crop_files)

crop_names <- sub("^GAEZAct2015_Production_", "", crop_names_raw)
crop_names <- sub("_Irrigated\\.tif$", "", crop_names)
crop_names <- tolower(crop_names)
crop_names <- gsub("\\\\.", "_", crop_names)
crop_names <- make.names(crop_names, unique = TRUE)

```

```

crop_names
# e.g. "banana", "barley", "cassava", ..., "yamsandotherroots"

# READ AS SPATRASTER STACK -----

r_crops <- rast(crop_files)
names(r_crops) <- crop_names

r_crops

# Convert to wide DT -----

gaze_dt_native <- as.data.table(as.data.frame(
  r_crops, xy = TRUE, cells = FALSE, na.rm = FALSE))

setnames(gaze_dt_native, c("x", "y"), c("lon", "lat"))

# Treat no-data as zero production -----

crop_cols <- setdiff(names(gaze_dt_native), c("lon", "lat"))

for (j in crop_cols) {
  set(gaze_dt_native, which(is.na(gaze_dt_native[[j]])), j, 0)
}

# Paths to our irrigated area stacks -----

template_02_path <- "./mha_stack_ll_02_aligned_NEAR.tif"
template_04_path <- "./mha_stack_ll_04_aligned_NEAR.tif"
template_10_path <- "./mha_stack_ll_10_aligned_NEAR.tif"

# CREATE FILES -----

# 0.2°-----

gaze_crops_02_aligned <- regrid_gaze_to_template(dt_native = gaze_dt_native,
  res = 0.2,
  template_path = template_02_path)

gaze_crops_02_aligned <- add_country_continent(gaze_crops_02_aligned, countries_sf,
  chunk_size = 200000L) %>%
  .[!is.na(country)]

```

Warning: Some values were not matched unambiguously: Fr. Polynesia, Fr. S. Antarctic Lands,

Warning: Some values were not matched unambiguously: Fr. Polynesia, Fr. S. Antarctic Lands,

```
# 0.4° -----

gaez_crops_04_aligned <- regrid_gaez_to_template(dt_native = gaez_dt_native,
                                                res = 0.4,
                                                template_path = template_04_path)

gaez_crops_04_aligned <- add_country_continent(gaez_crops_04_aligned, countries_sf,
                                                chunk_size = 200000L) %>%
  .[!is.na(country)]
```

Warning: Some values were not matched unambiguously: Fr. Polynesia, Fr. S. Antarctic Lands,

Warning: Some values were not matched unambiguously: Fr. Polynesia, Fr. S. Antarctic Lands,

```
# 1° -----

gaez_crops_1_aligned <- regrid_gaez_to_template(dt_native = gaez_dt_native,
                                                res = 1.0,
                                                template_path = template_1_path)

gaez_crops_1_aligned <- add_country_continent(gaez_crops_1_aligned, countries_sf,
                                                chunk_size = 200000L) %>%
  .[!is.na(country)]
```

Warning: Some values were not matched unambiguously: Fr. Polynesia, Fr. S. Antarctic Lands,

Warning: Some values were not matched unambiguously: Fr. Polynesia, Fr. S. Antarctic Lands,

```
# EXPORT FILES -----

fwrite(gaez_crops_02_aligned, "./datasets/crops/gaez2015_crops_irrigated_0p2deg_aligned.csv")
fwrite(gaez_crops_04_aligned, "./datasets/crops/gaez2015_crops_irrigated_0p4deg_aligned.csv")
fwrite(gaez_crops_1_aligned,  "./datasets/crops/gaez2015_crops_irrigated_1deg_aligned.csv")
```

```
#####
##### END RUNS #####
#####
```

1 Session information

```
# SESSION INFORMATION #####
```

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.3.1
##
## Matrix products: default
## BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/London
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] countrycode_1.6.1   sf_1.1-0      rnaturalearth_1.1.0
## [4] here_1.0.2          magrittr_2.0.4  terra_1.8-93
## [7] data.table_1.18.2.1
##
## loaded via a namespace (and not attached):
## [1] cli_3.6.5          knitr_1.51      rlang_1.1.7      xfun_0.55
## [5] otel_0.2.0         DBI_1.3.0       KernSmooth_2.23-26 rprojroot_2.1.1
## [9] htmltools_0.5.9    e1071_1.7-17    rmarkdown_2.30    grid_4.5.2
## [13] evaluate_1.0.5     classInt_0.4-11 fastmap_1.2.0     yaml_2.3.12
## [17] compiler_4.5.2     codetools_0.2-20 Rcpp_1.1.1        rstudioapi_0.17.1
## [21] digest_0.6.39      class_7.3-23    tools_4.5.2       proxy_0.4-29
## [25] units_1.0-0
```

```
## Return the machine CPU -----
```

```
cat("Machine:      "); print(benchmarkme::get_cpu()$model_name)
```

```
## Machine:
```

```
## [1] "Apple M1 Max"
```

```
## Return number of true cores -----
```

```
cat("Num cores:    "); print(parallel::detectCores(logical = FALSE))
```

```
## Num cores:
```

```
## [1] 10
```

```
## Return number of threads -----
```

```
cat("Num threads: "); print(parallel::detectCores(logical = FALSE))
```

```
## Num threads:
```

```
## [1] 10
```